

Name: _____

ANSWERS

Date: _____



Year 9 Mathematics

Test 1, 2015

Topic – Rates, Graphs, Indices



An Independent Public School

= _____ %

Total Time: 60 minutes

Total Reading: 5 minutes

Total Working: 55 minutes

Weighting: _____ % of the year.

Equipment:

 $\frac{1}{2}$ page notes (A4 one side), Scientific Calculator

SECTION 1: CALCULATOR FREE

Time:	38 minutes	Marks for Section 1:	36
Reading:	2 minutes	Equipment Allowed:	Nil
Working:	36 minutes		

1. [4 marks: 1, 1, 1, 1]
Simplify each of the following

a) $a^3 \times a^6$
 $= a^{3+6}$
 $= a^9$

b) $5m^4 \times 7m^5$
 $= 35m^{4+5}$
 $= 35m^9$

c) $2xy^5 \times x^3y^2$
 $= 2x^4y^7$

d) $8d^7e^3 \times d^2 \times 3de$
 $= 24d^{7+2+1}e^{3+1}$
 $= 24d^{10}e^4$

2. [4 marks: 1, 1, 1, 1]
Simplify each of the following

a) $\frac{y^6}{y^2} = y^{6-2} = y^4$

b) $\frac{21a^8}{3a^6} = 7a^{8-6} = 7a^2$

c) $\frac{4c^{11}}{24c^4} = \frac{1c^{11-4}}{6} = \frac{c^7}{6}$

d) $\frac{36k^{15}}{15k^8} = \frac{12k^{15-8}}{5} = \frac{12k^7}{5}$

3. [8 marks: 2, 2, 1, 3]

Simplify each of the following

a) $\frac{4p^3 \times 8p^6}{2p^2}$

$$= \frac{32p^{3+6}}{2p^2}$$

$$= \frac{32p^9}{2p^2} = 16p^{9-2}$$

$$= 16p^7$$

c) $\frac{40m^9n^{14}}{8m^6n^8}$

$$= 5m^{9-6}n^{14-8}$$

$$= 5m^3n^6$$

b) $\frac{3f^4 \times 15g^6}{5f \times g^2}$

$$= \frac{45f^4g^6}{5fg^2}$$

$$= 9f^{4-1}g^{6-2}$$

$$= 9f^3g^4$$

d) $\frac{3a^3 \times 4b^5 \times c^7}{5a \times b^5 \times 6c^4}$

$$= \frac{12a^3b^5c^7}{30ab^5c^4}$$

$$= \frac{2a^{3-1}b^{5-5}c^{7-4}}{5}$$

$$= \frac{2a^2b^0c^3}{5}$$

$$= \frac{2a^2c^3}{5}$$

4. [4 marks: 1, 1, 1, 1]

Simplify each of the following

a) $(f^3)^{11}$

$$= f^{33}$$

b) $(m^5n^2)^6$

$$= m^{30}n^{12}$$

c) $(3x^4y^3)^3$

$$= 3^3x^{12}y^9$$

$$= 27x^{12}y^9$$

d) $(p^3)^2 \times (q^5)^4$

$$= p^6 \times q^{20}$$

5. [4 marks: 2, 2]

Simplify each of the following

a) $\left(\frac{2d^5}{e^7}\right)^3$

$$= \frac{2^3 d^{15}}{e^{21}}$$

$$= \frac{8d^{15}}{e^{21}}$$

b) $\left(\frac{-2a^2b^9}{ab^7}\right)^4$

$$= \frac{(-2)^4 a^8 b^{36}}{a^4 b^{28}}$$

$$= 16a^{8-4}b^{36-28}$$

$$= 16a^4b^8$$

6. [2 marks: 1, 1]

Write each of the following using a negative index

a) $\frac{1}{7}$

$$= 7^{-1}$$

b) $\frac{1}{x}$

$$= x^{-1}$$

7. [5 marks: 1, 1, 1, 2]

Use index laws to simplify each of the following, express your answer with **positive indices**.

a) $p^2 \times p^{-5}$

$$= p^{2+(-5)}$$

$$= p^{-3}$$

$$= \frac{1}{p^3}$$

b) $\frac{24w^3}{16w^7}$

$$= \frac{3w^{3-7}}{2}$$

$$= \frac{3w^{-4}}{2}$$

$$= \frac{3}{2w^4}$$

c) $7a^4b \times 3a^{-6}b^3$

$$= 21a^{4+(-6)}b^{1+3}$$

$$= 21a^{-2}b^4$$

$$= \frac{21b^4}{a^2}$$

d) $18x^{-4}y^9 \div 9x^2y^3$

$$= 2x^{-4-2}y^{9-3}$$

$$= 2x^{-6}y^6$$

$$= \frac{2y^6}{x^6}$$

8. [5 marks: 1, 1, 1, 2]

Use index laws to simplify each of the following, express your answer with **positive indices**.

a) $\sqrt{9m^2}$

$$\begin{aligned} &= \sqrt{9} \times \sqrt{m^2} \\ &= 3 \times (m^2)^{1/2} \\ &= 3m \end{aligned}$$

b) $\sqrt{49t^4}$

$$\begin{aligned} &= \sqrt{49} \times \sqrt{t^4} \\ &= 7 \times (t^4)^{1/2} \\ &= 7t^2 \end{aligned}$$

c) $\sqrt[3]{x^3}$

$$\begin{aligned} &= (x^3)^{1/3} \\ &= x \end{aligned}$$

d) $\sqrt[3]{125d^6e^3}$

$$\begin{aligned} &= \sqrt[3]{125} \times \sqrt[3]{d^6} \times \sqrt[3]{e^3} \\ &= 5 \times (d^6)^{1/3} \times (e^3)^{1/3} \\ &= 5d^2e \end{aligned}$$

~ END OF TEST SECTION 1 ~

Name: _____

Date: _____

SECTION 2: CALCULATOR ASSUMED

Time:	27 minutes	Marks for Section 2:	35
Reading:	1 minutes	Equipment Allowed:	$\frac{1}{2}$ page notes (A4 one side), Scientific calculator
Working:	26 minutes		

9. [5 marks: 2, 2, 1]

 y is directly proportional to x , and $y = 24$ when $x = 6$.a) Find k , the constant of proportionality and the rule relating x and y

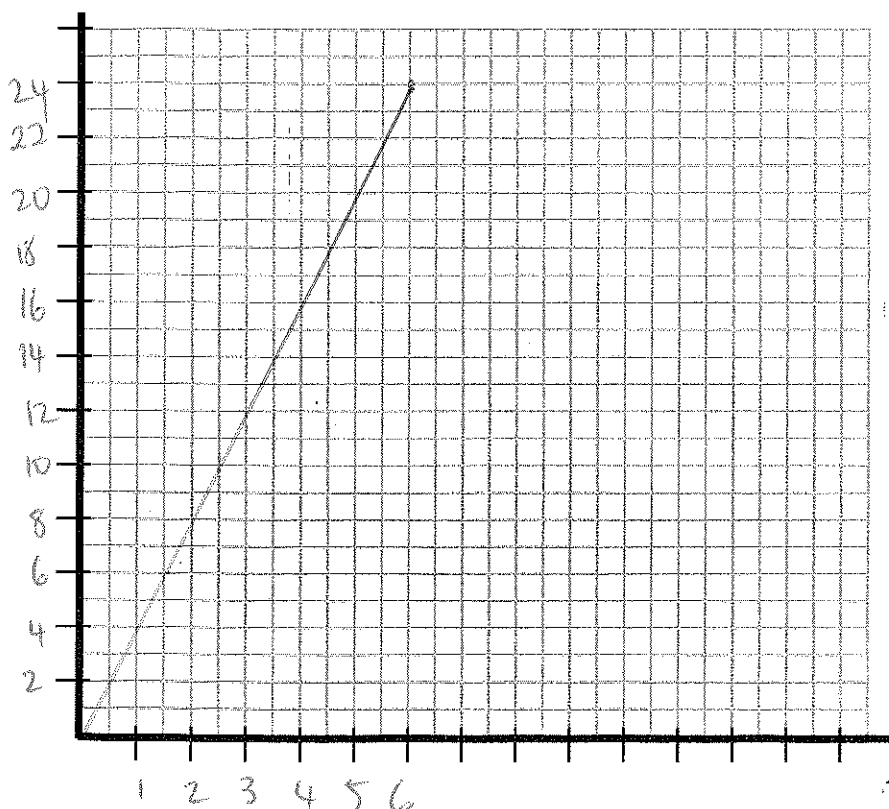
$$y = kx$$

$$k = \frac{y}{x}$$

$$k = \frac{24}{6}$$

$$= 4$$

$$\therefore y = 4x$$

b) Plot the graph relating x and y . (Hint: you may want to create a table of values)c) Find the value of y when $x = 14$.

$$y = 4x$$

$$y = 4 \times 14$$

$$= 56$$

10. [4 marks: 1, 2, 1]

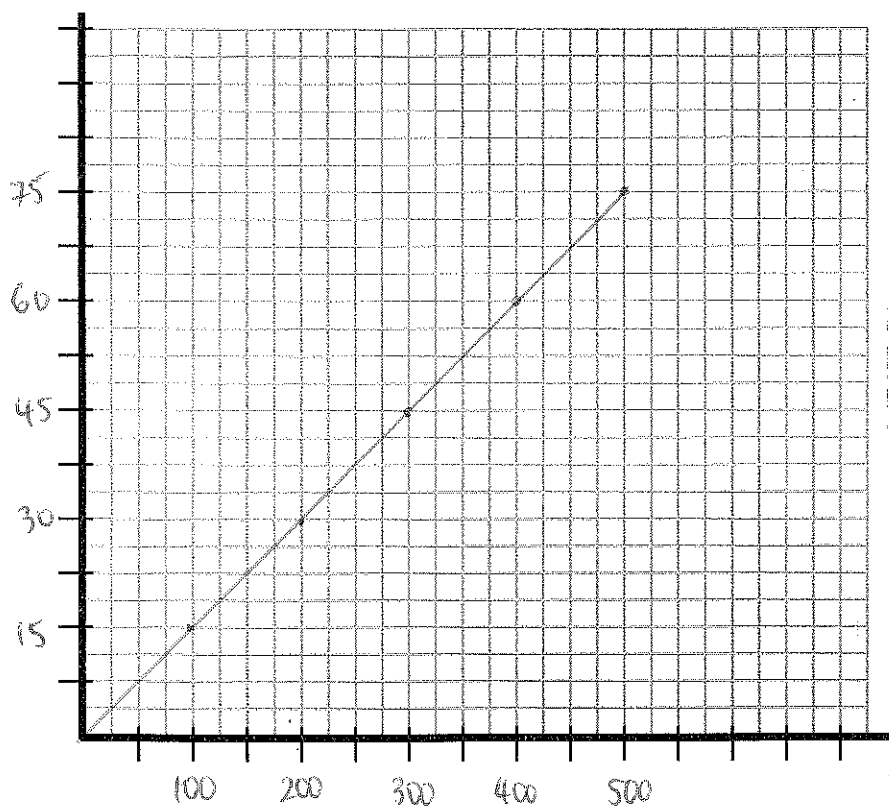
An employee gets \$0.15 for each leaflet delivered. Use x as the number of leaflets delivered and y as the total money they earn in a week.

a) Determine the rule relating y and x

$$y = 0.15x$$

x	100	200	300	400	500
y	15	30	45	60	75

b) Plot the graph relating y and x (for values of x in units of 100)



c) Determine how much money the person earns if they deliver 1245 leaflets.

$$\begin{aligned} y &= 0.15 \times 1245 \\ &= \$186.75 \end{aligned}$$

11. [4 marks: 2, 2]

Find k , the constant of proportionality if:

- a) a is directly proportional to b , and $a = 275$ when $b = 25$

$$\begin{aligned} k &= \frac{a}{b} \\ &= \frac{275}{25} \\ &= 11 \end{aligned}$$

- b) m is directly proportional to n , and $m = 42$ when $n = 28$

$$\begin{aligned} k &= \frac{m}{n} \\ &= \frac{42}{28} \\ &= \frac{3}{2} \end{aligned}$$

12. [4 marks: 2, 1, 1]

8 calculators cost \$116. Let c = cost of calculator, and let n = number of calculators

- a) Write a rule for the relationship between c and n

$$\$116 \div 8 = \$14.50$$

$$C = \$14.50n$$

- b) How much would 10 calculators cost in total?

$$\begin{aligned} C &= 14.50 \times 10 \\ &= \$145 \end{aligned}$$

- c) How many calculators could I buy for \$200

$$\begin{aligned} C &= 14.5n \\ 200 &= 14.5n \\ \frac{200}{14.5} &= n \end{aligned}$$

$$n = 13 \text{ calculators}$$

13. [4 marks: 2, 2]

a varies inversely as b , and $a = 8$ when $b = 10$

a) Calculate k , the constant of proportionality and the rule relating a and b

$$y = \frac{k}{x}$$

$$\therefore k = x \times y$$

$$k = a \times b$$

$$= 8 \times 10$$

$$= 80$$

$$\therefore a = \frac{80}{b}$$

b) Find b when $a = 17$, correct to 2 decimal places.

$$a = \frac{80}{b}$$

$$17 = \frac{80}{b}$$

$$17 \times b = 80$$

$$b = \frac{80}{17}$$

$$= 4.71 \text{ (2 d.p.)}$$

14. [5 marks: 2, 2, 1]

y is inversely proportional to x , and $y = 4$ when $x = 6$.

a) Find k , the constant of proportionality and the rule relating y and x

$$k = x \times y$$

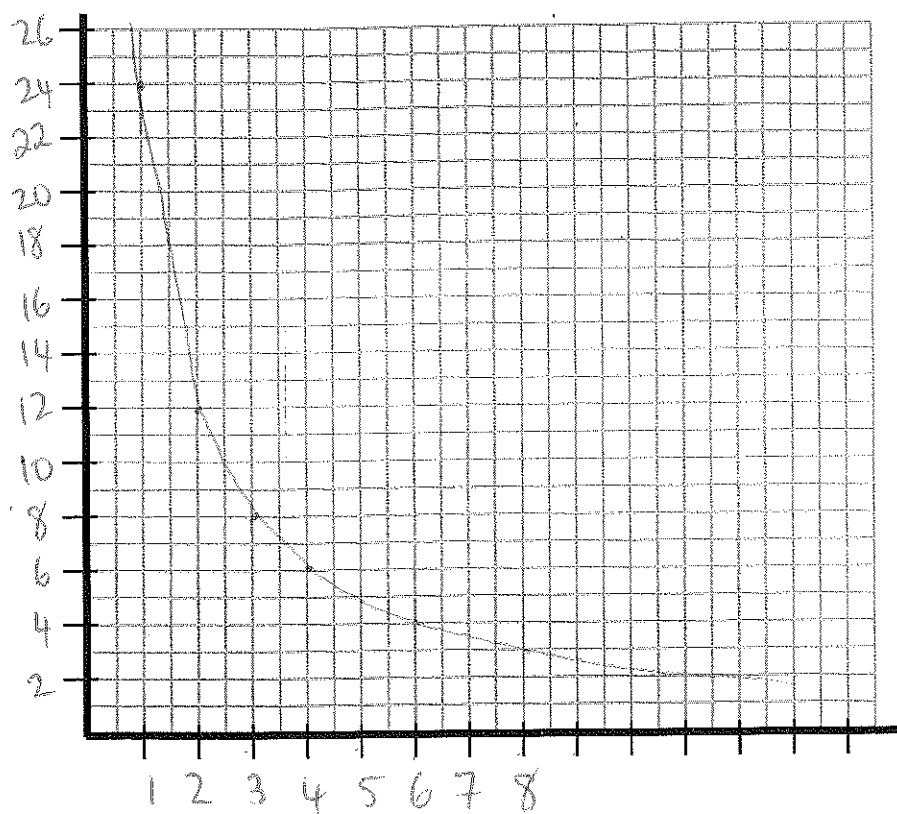
$$k = 4 \times 6$$

$$= 24$$

$$\therefore y = \frac{24}{x}$$

b) Plot the graph relating y and x (for values of x that are factors of k up to 12)

$$y = \frac{24}{x}$$



x	1	2	3	4	6	8
y	24	12	8	6	4	3

c) Find the value of y when $x = 10$

$$y = \frac{24}{x}$$

$$y = \frac{24}{10} = 2.4$$